UNIVERSITY AT ALBANY
State University of New York

College of Arts and Sciences
Department of Mathematics and Statistics

Colloquium

Matt Zaremsky

On the Boone-Higman Conjecture and Related Problems

Friday, November 20, 2026
3:00 p.m. Massry B010
(tea & coffee at 2:40 p.m.)

Abstract. The Boone-Higman conjecture, posed in the early 1970's, predicts that every finitely generated group with solvable word problem embeds in a finitely presented simple group. A stronger form of the conjecture asks that the simple group even be of type F_∞. I will discuss the progress on these and related conjectures that I and various coauthors (e.g., Belk, Bleak, Fournier-Facio, Hyde, Matucci) made over the course of the 2020's, and the recent claimed proof by OpenAI building off our work.